

Question Paper

Exam Date & Time: 29-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

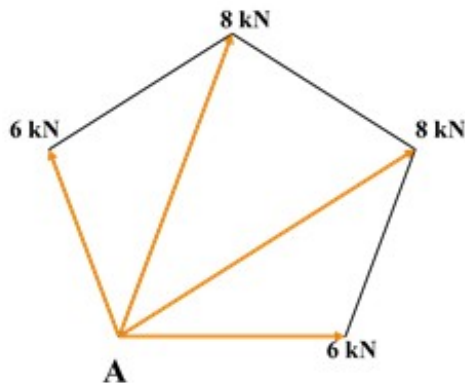
FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2025
SUBJECT: CIV 1003/CIV_1003 ENGINEERING MECHANICS AND SMART BUILDINGS

Marks: 50

Duration: 180 mins.

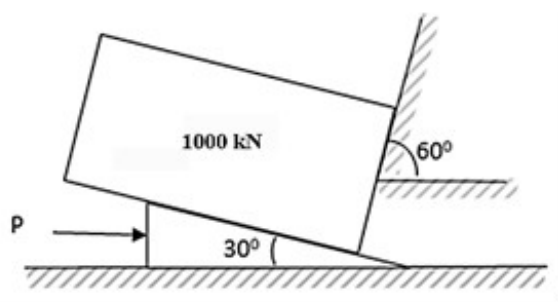
Answer all the questions.

- 1A) A regular pentagon has its vertices on a circle. From vertex A four forces are applied towards the other four vertices with magnitudes 6 kN, 8 kN, 8 kN, and 6 kN, respectively as shown in the figure. Determine the resultant force. (5)



Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Determine the force P required to impend motion of 1000 kN block upwards. Consider the coefficient of limiting friction as 0.25 for all contact surfaces. Neglect the weight of the wedge. (5)



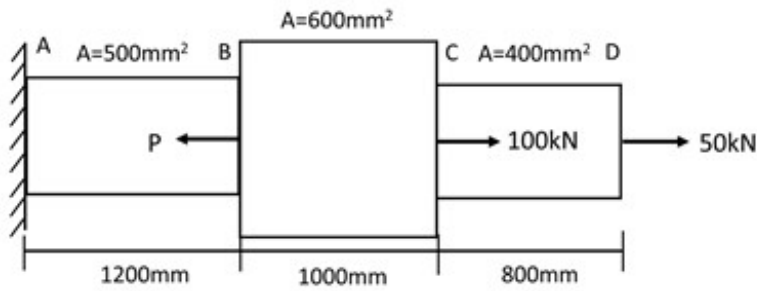
Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) Explain how access control and fire alarm systems together contribute to building safety and occupant security in a smart building environment. (5)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2B) Determine the magnitude of load P required to establish the condition under which there is no net (5)

change in the total length of the bar ABCD shown in the figure. Assume $E = 200 \text{ GPa}$.



Course Outcome: "CO4" / Cognitive Level: "Application"

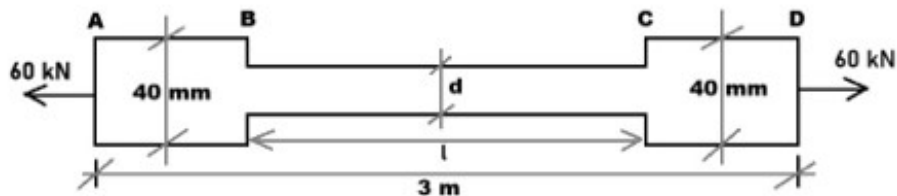
- 3A) Derive an expression for elongation of a tapered bar of rectangular cross-section subjected to an axial tensile load "W". The bar has varying depths from d_1 to d_2 ($d_1 < d_2$) and uniform thickness 't' over length L. (5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 3B) An aluminium block of size 300 mm (length) x 100 mm (width) x 50 mm (height) is subjected to a tensile force of 150 kN along its length and a compressive force of 200 kN along its width. Determine the change in its length, width, and height. Assume the modulus of elasticity (E) for aluminium as 70 GPa and Poisson's ratio as 0.33. (5)

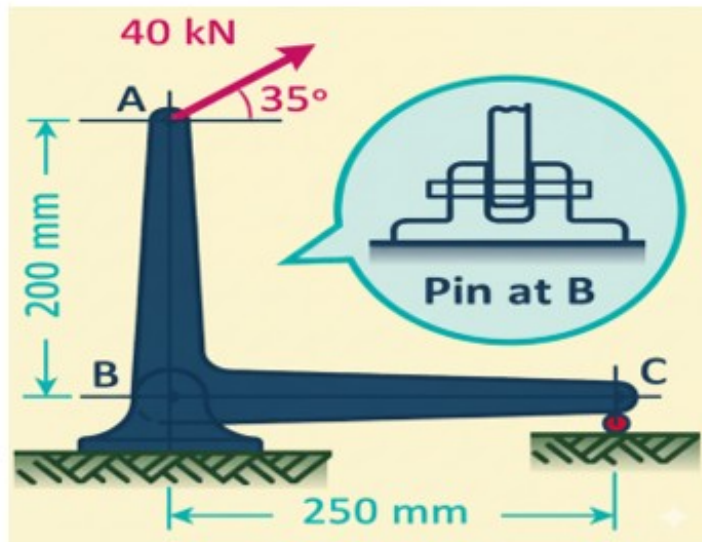
Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) A circular bar ABCD, 3m long is subjected to a tensile force of 60kN as shown in the figure. If the stress in the middle portion BC is not to exceed 175 MPa, then what should be its diameter? Also find the length of the middle portion, if the total extension of the bar should not exceed by 3.25 mm. Take E as 100 GPa. (5)



Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) Compute the shearing stress in the pin/hinge at B for the member supported as shown in Figure. The pin/hinge diameter is 20 mm. (5)



Course Outcome: "CO4" / Cognitive Level: "Application"

5A) Explain is the necessity of instrument calibration? List any six key factors. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

5B) Explain working Mechanism of
i) Force Transducer.
ii) Distance sensor. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		CIV_1003:ENGINEERING MECHANICS AND SMART BUILDINGS	
#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	5
4	4	CO 4	25
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		CIV_1003:ENGINEERING MECHANICS AND SMART BUILDINGS	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	8	40(80%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)

Total	10	50 (100%)
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